

Physiologically-Triggered, Receptivity-Aware Hydration Prompting

A Low-Cost Wearable System and Micro-Randomized Evaluation

Project Document — Version 1.0

September 2026

*Prepared for internal review. Sections 11 and 12 contain verification tasks
that must be completed before external circulation.*

Contents

Abstract

Hydration reminder systems, both commercial and research-grade, almost universally trigger on a clock or on a fixed daily volume goal. This is a design choice made for convenience rather than physiology: the moment a person is told to drink bears no relation to the moment their body is actually accruing a fluid deficit, and no relation to whether they are in a position to act. The consequence is prompts that arrive when they cannot be acted on, habituation, and eventual disengagement.

We propose a wearable system that triggers hydration prompts on **estimated physiological fluid deficit** rather than elapsed time, and times those prompts using a **learned receptivity model** constrained by a **need-based safety override** that receptivity is structurally incapable of suppressing. Fluid deficit is estimated by a grey-box model: a published whole-body sweat-rate heat-balance equation whose coefficients are individualized online from supervised calibration sessions referenced against the scale method, combined with objective intake measured by a load-cell-instrumented bottle. Sensing is performed by a sub-Rs. 3,000 ESP32 wearable carrying PPG, inertial, skin-temperature, and ambient temperature/humidity sensors.

The system will be evaluated by **micro-randomized trial** (MRT) in a heat-exposed athletic population ($n = 10\text{--}12$, 14 days), randomizing the prompt decision at each eligible decision point. We report three planned contributions: (i) an ablation quantifying what low-cost physiological sensing adds to a validated activity-and-environment sweat-rate equation; (ii) the first estimate of the proximal causal effect of a hydration prompt on near-term drinking behaviour; and (iii) an empirical characterization of the divergence between adherence-maximizing and deficit-minimizing prompting policies — a failure mode we argue is intrinsic to feedback-driven health interventions and not specific to hydration.

We explicitly do not claim novelty in sweat-rate prediction, smart-bottle intake sensing, just-in-time reminder delivery, or micro-randomized trial methodology, all of which are established. The contribution lies in the trigger, the constraint structure, and the evaluation design.

Keywords: *wearable sensing; hydration; just-in-time adaptive intervention; micro-randomized trial; constrained bandit; ESP32; fluid balance*

1. Introduction

1.1 Motivation

Water regulates thermoregulation, cardiovascular function, cognition, and renal health. Fluid deficits of as little as 2% body mass measurably degrade physical and cognitive performance, and larger deficits carry clinical risk under heat stress. Despite this, sustaining adequate intake is a persistent behavioural failure, and the reason is well documented: thirst is a lagging indicator. By the time thirst is felt, a deficit already exists. People consistently report forgetting to drink and not feeling thirsty as their principal barriers.

This makes hydration an unusually clean target for technological support. The behaviour required is trivial — drink — and the failure is one of timing and salience, not capability or motivation. If a system can identify the right moment and deliver a prompt the person is able to act on, the behavioural distance from prompt to action is a few seconds.

1.2 The problem with existing systems

Essentially every deployed hydration reminder — commercial smart bottles, phone apps, and the research systems built on them — triggers on one of two rules:

1. Clock-based: remind every n minutes.
2. Volume-lapse: remind if a fixed hourly or daily intake target has not been met.

Neither rule contains any information about the user’s physiological state. A sedentary person in air conditioning and a footballer in 34 °C, 75% RH receive structurally identical treatment, differing only in the constant. Yet whole-body sweat rate varies several-fold between individuals at identical workload and environment, and several-fold again within an individual across conditions. A rule that ignores this is, at any given moment, either over-prompting (breeding habituation) or under-prompting (missing the moments that matter).

Neither rule contains any information about **receptivity** either — whether the user is in a position to act. A prompt delivered mid-sprint, mid-meeting, or with no water within reach is not merely wasted; it trains the user to dismiss prompts.

1.3 The trap in fixing receptivity naively

The obvious fix — learn from user responses and stop prompting when they do not respond — contains a failure mode that is dangerous in a health context. Non-response is not evidence of non-need. A system that optimizes for prompt compliance converges toward prompting the people and moments that were already easy, and away from the people and moments that are hard. In physical-activity applications this is merely inefficient; a missed walking prompt harms nobody. In hydration under heat stress, the moments of lowest receptivity (intense activity, high heat) frequently coincide with the moments of highest need.

We treat this not as a bug to patch but as the object of study. Our policy architecture separates **need** (physiological, never suppressible by behaviour) from **receptivity** (behavioural, governing timing and modality within a need-opened window), and we measure empirically where an adherence-optimizing policy and a deficit-minimizing policy disagree.

1.4 Scope and honest positioning

This project does **not** measure hydration status directly. No consumer-grade sensor does. We estimate a *fluid balance* — losses minus intake — using established physiology for the loss term and direct measurement for the intake term. The word “prediction” in this document refers to sweat rate and to short-horizon deficit projection, never to a diagnosis of dehydration.

2. Literature Review

2.1 Wearable hydration monitoring

A 2025 scoping review of wearable hydration-monitoring technologies surveyed sensor modalities, benchmarked accuracy, and reviewed commercial products [1]. Its conclusions bear directly on our positioning: wearable hydration devices provide real-time assessment but face persistent challenges in reliability, accuracy, and generalization across populations, and the review names multimodal systems integrating sensors with AI-driven analysis for personalized hydration

management as a future direction. Our system sits inside that named direction, which establishes relevance but simultaneously establishes that the general concept is not itself novel.

The review documents the sensor landscape. Bioimpedance approaches use surface electrodes to detect hydration-linked impedance change, typically validated on very small samples. Skin hydration sensors using silver-nanowire electrodes or thermistor-based measurement of volumetric water content are calibrated against commercial dermatological instruments; critically, these measure stratum corneum water content — a dermatological construct — not systemic fluid balance. Sweat electrochemical sensors measure sodium, potassium and pH, with prototypes demonstrated on athletes and machine-learning models predicting body water loss percentage from biomarkers including heart rate and sweat sodium concentration. That last approach is the closest existing work to a “predict hydration with ML” framing, and it depends on sweat electrolyte biosensing hardware well outside our cost envelope.

2.2 Sweat-rate prediction: a solved problem

This is the literature that determines what we can and cannot claim about our modelling layer.

Gonzalez et al. [2] developed corrected and new whole-body sweat-loss prediction equations from careful measurements in 101 volunteers producing over 500 observations, across ambient temperatures of 15–46 °C, water vapour pressures of 0.27–4.45 kPa, wind speeds of 0.4–2.5 m/s, varied clothing and equipment, metabolic rates from rest to ≤ 450 W/m², and durations of 2–8 hours. The resulting piecewise equation was cross-validated against independent data.

More recently, whole-body sweat rate prediction equations were developed and validated separately for indoor treadmill and cycle ergometer exercise [3] and for outdoor running and cycling [4], using the human heat balance equation rearranged to solve for required evaporative heat loss, with metabolic rate derived from simple activity inputs. These models were built and validated on separate participant groups across 20–40 °C dry-bulb and 14–60% relative humidity, at 40–85% of peak aerobic power. The authors report that the predictions provide excellent estimates of fluid loss but note that they **may require individual validation**. That caveat is the entire remaining opening: the physics is settled, and per-individual calibration is the acknowledged residual limitation.

Baker [5] reviews sweating-rate methodology and intra/inter-individual variability, confirming that the simplest and most accurate method for assessing whole-body sweat rate is change in body mass across exercise. Cheuvront et al. [6] characterize the limits of accurate sweat-loss prediction over prolonged exercise, and Maughan et al. [7] document the error sources in inferring hydration status from body-mass change — respiratory water loss, metabolic mass loss, incomplete drying. These define the error floor of our reference method and must be reported, not ignored. Field-condition validation continues: a 2025 study compared a commercial wearable sweat sensor and a sweat-rate calculator against the scale-based method in hot-dry (40 °C, 36% RH) and hot-humid (30 °C, 81% RH) conditions in 23 participants [8].

Implication for this project: we use published equations as our baseline, and our contribution at this layer is limited to online per-user coefficient fitting from low-cost wrist sensing. This is an incremental engineering delta and will be framed as such.

2.3 Fluid intake sensing

Smart bottles are mature and validated. A comparison of four commercial connected bottles against a high-resolution weight scale across 100 intake events per bottle characterized per-sip and cumulative error, finding load-cell-based designs balanced error across multiple sips well [9]. A separate study validated a smart bottle against a reference method during American football pre-season training, reporting no significant mean difference and Bland–Altman limits of agreement of –212 to 180 mL, corresponding to roughly $\pm 15\%$ at the individual level [10].

Load-cell instrumentation of a bottle base is therefore an established, validated technique, not a contribution. We adopt it because it converts intake from self-report to measurement, which is a precondition for every downstream claim in this project.

2.4 Just-in-time hydration interventions — the closest prior art

sipIT (Streeper and colleagues) is the system nearest to ours and must be engaged directly. sipIT combines a connected water bottle, a smartwatch clockface performing drinking-gesture detection, and a companion app, delivering just-in-time text-message reminders when a periodic fluid intake goal is not met [11]. Its feasibility study captured tens of thousands of drinking events over three months in kidney-stone patients, reported increased automaticity of fluid intake, and reduced the “lack of thirst” barrier. A reduced-cost variant using only the bottle and app produced a significant increase in 24-hour urine volume. The full intervention is now under evaluation in a randomized efficacy trial against usual care [11].

Related clinical work includes a randomized trial of a commercial smart bottle in stone formers [12], analysis of smart technology’s impact on adherence and fluid management in stone prevention [13], and a protocol for a two-arm RCT of an applet-based fluid intake reminder in post-operative urinary calculi patients [14].

Four properties characterize this entire body of work:

1. The trigger is behavioural — a lapse against a fixed volume goal — not physiological.
2. There is no receptivity model; prompts fire on lapse regardless of the user’s capacity to act.
3. Evaluation is parallel-arm RCT against usual care, which establishes whether the package works but cannot isolate the effect of a prompt at a given moment.
4. The populations (kidney stone patients, elderly, post-operative) face slow deficit accrual with no thermal challenge.

2.5 Just-in-time adaptive interventions and micro-randomized trials

The JITAI framework formalizes decision points, tailoring variables, intervention options, and decision rules [15]. The micro-randomized trial was developed specifically to support JITAI optimization by randomizing the intervention decision at each decision point, enabling unbiased estimation of proximal treatment effects within subject [16]. Sample-size methodology for MRTs in mHealth is established [17], and the design has been argued for as a general public-health tool [21].

Within this literature, the question of when to intervene has been studied directly. Liao et al. [18] address determining treatment timing and the cost of over-treatment, and the HeartSteps line of work developed reinforcement-learning personalization of activity prompts [19]. Reinforcement learning for personalizing intervention communication has also been evaluated in medication adherence [20].

Implication: the machinery of receptivity-aware intervention is not new. Its application to hydration is not present in the literature we located, and we found no micro-randomized trial of any hydration intervention.

3. Research Gaps

Synthesizing Section 2, five gaps are identified. G1 and G2 are minor; G3–G5 constitute the project’s contribution.

ID	Gap	Evidence from §2
G1	Published sweat-rate equations are validated at group level but acknowledge that individual validation may be required. No system fits individual coefficients online from continuous low-cost wrist sensing.	[3][4] state the individual-validation caveat explicitly
G2	The marginal value of consumer-grade PPG and skin temperature over activity-and-environment inputs alone has not been quantified for fluid-loss estimation.	Reviewed systems either use lab-grade sensing [8] or omit the comparison entirely
G3	All existing hydration prompting is triggered by clock or volume lapse. No system triggers on estimated physiological deficit.	sipIT [11], commercial bottles [9], applet RCT [14] — all goal-lapse triggered
G4	No hydration intervention models user receptivity, and no receptivity-aware health intervention in any domain formalizes a constraint preventing learned non-response from suppressing high-need alerts.	Receptivity and timing work exists [18][19] but in domains where a missed prompt is harmless
G5	No micro-randomized trial of a hydration intervention exists. The proximal causal effect of a hydration prompt on near-term drinking has never been estimated.	All located hydration trials are parallel-arm [11][12][14]

A sixth, methodological gap follows from G4 and generalizes beyond hydration:

G6: *The divergence between adherence-maximizing and need-minimizing policies in feedback-driven health interventions has been noted conceptually but never measured empirically in a domain where the two objectives materially conflict.*

4. Novelty Statement

4.1 Claimed contributions

C1 — Physiologically-triggered hydration prompting. Prompts are gated by a short-horizon projection of estimated fluid deficit derived from continuous physiological and environmental sensing, replacing the clock or volume-goal trigger universal in prior systems (addresses G3).

C2 — Constrained receptivity-aware policy. A two-model architecture in which a need estimator opens intervention windows and a learned receptivity model selects timing and modality within windows, subject to a hard physiological override that receptivity is structurally unable to suppress.

Formulated as a contextual bandit constrained on an accumulating physiological state variable that the agent’s own inaction worsens — a constraint structure distinct from the burden or frequency caps used in existing mHealth bandit work (addresses G4).

C3 — First micro-randomized evaluation of hydration prompting, yielding an unbiased proximal effect estimate and an off-policy-evaluable dataset (addresses G5).

C4 — Empirical characterization of adherence/need policy divergence, measuring where and how much a compliance-optimizing policy and a deficit-minimizing policy disagree, stratified by activity state and time of day (addresses G6). We regard this as the contribution with the broadest reach, as it concerns JITAI design generally and uses hydration as a testbed.

C5 — Ablation of low-cost physiological sensing value against a validated activity-and-environment baseline, reported regardless of direction (addresses G1, G2).

4.2 Explicitly disclaimed

For the avoidance of reviewer objection, the following are not claimed as novel and are cited as prior art in Section 2: sweat-rate prediction from activity and environment [2][3][4]; load-cell fluid intake sensing [9][10]; just-in-time hydration reminders combining bottle and wearable [11]; micro-randomized trial methodology [16][17]; receptivity modelling in mHealth [18][19]; and low-cost ESP32 multimodal sensing platforms generally.

4.3 Assessed novelty by component

Component	Assessment	Basis
Multimodal low-cost sensing platform	Low	Commodity
Sweat-rate estimation	Low	Solved [2][3][4]
Online per-user coefficient fitting	Moderate	Fills an acknowledged limitation
Instrumented bottle + prompting	Low	sipIT [11]
Physiological deficit trigger (C1)	Moderate–High	No precedent located
Constrained receptivity policy (C2)	High	Constraint structure appears novel
MRT for hydration (C3)	High	First application
Policy divergence measurement (C4)	High	Generalizes beyond domain

5. Research Questions and Hypotheses

ID	Question	Hypothesis	Primary measure
RQ1	Does adding PPG and skin temperature improve fluid-loss estimation over an activity-and-environment baseline?	H1: modest improvement; possibly null	RMSE; Bland–Altman LoA vs. scale method
RQ2	Do per-user coefficients beat	H2: yes; plateau by 3–4	Error vs. n sessions (learning

ID	Question	Hypothesis	Primary measure
	population defaults, and after how many calibration sessions?	sessions	curve)
RQ3	What is the proximal effect of a prompt on drinking within 30 min?	H3: positive, moderate	$P(\text{drink} \leq 30 \text{ min} \mid \text{prompt}) - P(\text{drink} \leq 30 \text{ min} \mid \text{no prompt})$
RQ4	Does receptivity-aware timing reduce time-in-deficit vs. lapse-contingent prompting?	H4: yes	Minutes/day above deficit threshold
RQ5	Where do adherence-maximizing and deficit-minimizing policies diverge?	H5: divergence concentrates in high-activity, high-heat contexts	Policy disagreement rate, stratified

RQ1 and RQ2 can return null results and remain publishable: “low-cost physiological sensing adds little over validated equations” is a useful negative finding, and the study is structured so that the paper survives either direction.

6. System Architecture

6.1 Block diagram

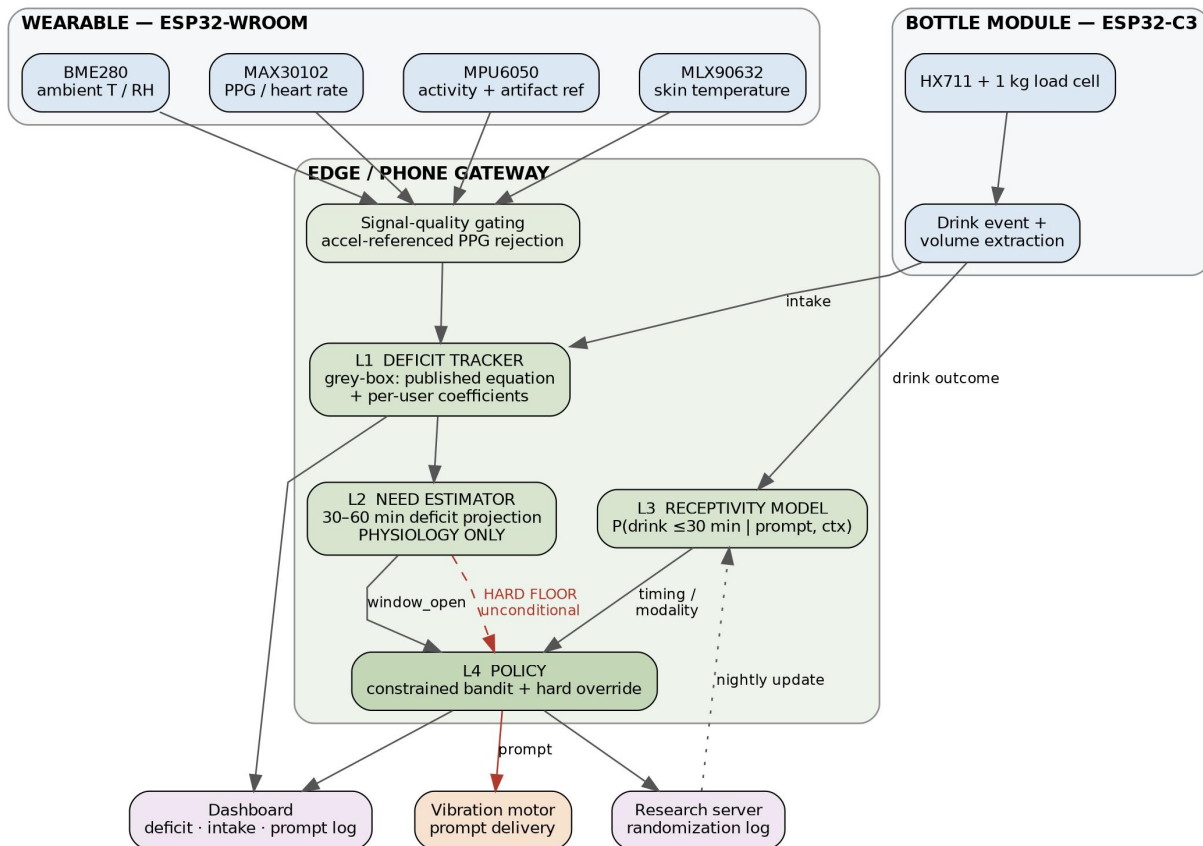


Figure 1. End-to-end system architecture. Note the two paths from L2 into L4: the ordinary `window_open` signal, and the dashed hard-floor override which bypasses the receptivity model entirely.

6.2 Layer specifications

L1 — Deficit tracker (grey-box)

$$\text{deficit}(t) = \int [\text{sweat_rate}(t) + \text{insensible_rate}] dt - \int \text{intake}(t) dt$$

- `insensible_rate` — fixed literature constant; not modelled.
- `sweat_rate(t)` — heat-balance equation form per [3][4], with per-user scaling and intercept coefficients fitted from calibration.
- Cold start uses population defaults. After ≥ 3 calibration sessions, individual coefficients replace them.
- Output: deficit in mL and as % body mass.

Four variants are implemented for the RQ1 ablation, selectable by configuration flag:

Variant	Inputs
A — Context-only	activity intensity, ambient T, RH
B — Physiology-only	HR, skin temperature
C — Full	all of the above
D — Fixed-rate baseline	single per-person sweat rate from one session, temperature-scaled

Variant D is the honest baseline. If C does not beat D by a margin that matters practically, that is the finding and it will be reported as such.

L2 — Need estimator

Projects deficit 30–60 min forward from current activity and environment. Outputs need $\in [0,1]$ and a binary `window_open` flag. **Behavioural features are structurally excluded from this layer.**

L3 — Receptivity model

$P(\text{drink} \leq 30 \text{ min} \mid \text{prompt, context})$. Features: time of day, activity state, minutes since last prompt, prompt count today, bottle mass (availability proxy), day of week. Trained online from bottle-verified drink events. **Logistic regression or small gradient-boosted trees** — with approximately 1,900 decision points, higher-capacity models will overfit and invite justified reviewer objection.

L4 — Policy

```

if projected_deficit >= HARD_FLOOR:      # ~2% body mass
    prompt immediately, escalate modality  # unconditional override
elif window_open:
    if receptivity >= theta and cooldown elapsed:
        prompt

```

```

else:
    defer; re-evaluate next decision point
    if window open >= MAX_DEFER: prompt regardless
else:
    no prompt

```

Design invariant: low receptivity may delay a need-driven prompt; it may never cancel one. HARD_FLOOR and MAX_DEFER are fixed constants, not learned parameters.

Learning objective: minimize cumulative time-in-deficit, not maximize prompt response rate. RQ5 exists precisely because these objectives diverge.

6.3 Design decisions and rationale

Decision	Rationale
GSR / EDA removed	Wrist EDA is dominated by sympathetic arousal (stress, caffeine, heat). The skin-hydration literature it borrows credibility from measures stratum corneum water content [1], a different construct from systemic fluid balance.
Skin temperature added	Thermal strain is the most hydration-relevant physiological channel; ambient temperature alone leaves it unmeasured.
Bottle load cell added	Converts intake from self-report to measurement. Precondition for the deficit trace, the ground truth, and the MRT outcome.
Athletic population	Personal bottle use is habitual, making intake observable. Communal drinking sources in occupational settings render intake invisible, which is why that population was ruled out.
MRT decision points in non-training hours	During supervised training, coaches enforce drinking; deficit is small and the effect ceiling is near zero. Training generates calibration data; the remainder of the day generates the trial.
Minimal dashboard	A five-panel dashboard is not graded and consumes weeks. Deficit trace, intake log, prompt log only.

7. Implementation Plan

7.1 Bill of materials

Item	Qty	Est. Rs.
ESP32-WROOM dev module (wearable)	2	700
ESP32-C3 mini (bottle)	2	500
MAX30102 breakout	2	400
MPU6050	2	200
MLX90632 (DS18B20 fallback @ Rs. 80)	2	1,400
BME280	2	500

Item	Qty	Est. Rs.
Coin vibration motor + driver	2	200
HX711 + 1 kg load cell	2	700
LiPo 500 mAh + TP4056	4	800
Strap, enclosure, 3D printing	—	800
Per unit (wearable + bottle base)		~2,600
Precision scale, 20 g resolution, 150 kg capacity	1	2,500
Handheld refractometer (USG cross-check)	1	2,800
Project total		~11,000

Build two complete units minimum. Single-unit failure mid-protocol costs weeks.

7.2 Measurement protocol

Calibration sessions (supervised). Five per subject, varied by intensity (moderate / hard) and time of day (cooler morning / hotter afternoon).

1. Void bladder. Record nude body mass to 20 g. Optional USG via refractometer.
2. Fit wearable. Weigh bottle at known fill.
3. Exercise 45–60 min, continuous 1 Hz logging, ad libitum drinking from the instrumented bottle only.
4. No urination during session; if unavoidable, collect and weigh.
5. Towel dry thoroughly. Re-record nude body mass. Re-weigh bottle.

$$\text{WBSL} = \text{mass_pre} - (\text{mass_post} - \text{intake} + \text{urine_output})$$

This is the accepted reference method [5]. Known error sources — respiratory water loss, metabolic mass loss, incomplete drying [6][7] — are reported, not ignored.

Free-living deployment (MRT). 14 days per subject; wearable and bottle carried during waking hours; decision points every 30 min during non-training waking hours subject to window_open; prompt randomized with $p = 0.5$ at each eligible decision point. Hard-floor prompts are not randomized, are excluded from RQ3, and are logged separately.

7.3 Population

University field-sport athletes or a running club, $n = 10\text{--}12$ (over-recruit to 14).

- Inclusion: 18+, trains $\geq 3\times/\text{week}$, no cardiac or renal condition, not taking diuretics.
- Exclusion: pregnancy, current illness, medication affecting thermoregulation or fluid balance.

Power. MRTs are within-subject. 10 subjects $\times$ 14 days $\times$ ~14 eligible decision points per day $\approx$ 1,900 randomized decision points. A formal power calculation following Liao et al. [17] must be reported; adequacy is not to be asserted.

7.4 Analysis plan

RQ	Method
RQ1, RQ2	Leave-one-session-out cross-validation. RMSE and Bland–Altman limits of agreement vs. scale method, per variant A/B/C/D. Error-vs-sessions learning curve. Report Bland–Altman, not R^2 alone — correlation hides bias, and bias accumulates in an integrator.
RQ3	GEE / weighted centred least squares for a proximal binary outcome, clustered by subject, per [16].
RQ4	Time-in-deficit per day, adaptive vs. simulated lapse-contingent policy on identical logged data.
RQ5	Off-policy evaluation on randomized data: fraction of decision points where compliance-maximizing and deficit-minimizing policies disagree, stratified by activity state and time of day.

Pre-register RQ3 and RQ4 (OSF or AsPredicted) before deployment. Because the data are randomized, off-policy evaluation of policies never deployed is unbiased — this multiplies the yield of a small cohort and is the reason to insist on randomization rather than deploying a single adaptive policy.

7.5 Timeline

Sixteen weeks nominal; treat as twenty with realistic slack.

Wk	Work	Gate
1	Ethics submission filed, including a data-sharing clause. Order parts.	Ethics filed
2–3	Wearable firmware: drivers, 1 Hz logging, BLE. Bottle load cell calibrated to ± 10 mL.	Bench-validated logging
4	PPG signal-quality gating; characterize artifact rejection during running.	Known % usable PPG
5	L1 with published equations and population defaults. Dashboard skeleton.	Deficit trace end-to-end
6	Pilot calibration protocol ($n=2$, one session each). Fix what breaks.	Protocol validated
7	Recruitment, consent, baseline. Approval must be in hand.	Ethics approved
8–10	Calibration sessions, five per subject.	~55 sessions logged
11	Fit per-user coefficients. RQ1/RQ2 ablation. Initial receptivity model.	GO / NO-GO on MRT
12–13	MRT deployment days 1–14. Daily data-integrity checks.	Randomization log clean
14	MRT completion, exit interviews, hardware return.	Dataset frozen
15	Analysis: RQ3, RQ4, RQ5.	Results
16	Writing.	Draft submitted

Week 11 is a genuine kill gate. If sweat-loss prediction RMSE exceeds approximately 25%, the need model has no physiological basis and an MRT built on it produces meaningless results. Fall back to Section 9 rather than proceed.

7.6 Risk register

Risk	Lk.	Impact	Mitigation
Ethics approval delayed past week 7	High	Fatal	File week 1; written fallback (§9)
PPG unusable during activity	High	Loses key input	Accel-referenced gating; report usable fraction as a finding; skin temp carries signal if HR fails
Athletes already hydrate well in training	High	Kills RQ3	Decision points in non-training hours only
Bottle bypassed (fountain, canteen, teammate)	Medium	Corrupts intake	In-app manual log; report bypass rate; pre-set exclusion threshold
Attrition below $n = 8$	Medium	Underpowered	Over-recruit to 14; minimize burden
Sweat-rate error too large	Medium	Undermines L2	Week 11 gate
Hardware failure mid-deployment	Medium	Data loss	Two spare units; local SD logging so BLE dropouts do not lose data
Ablation returns null	Medium	Not a risk	Publishable negative finding; pre-registration makes it a result, not a failure

8. Deliverables

1. Wearable and bottle module, two complete units; firmware in a public repository.
2. Calibration protocol document at methods-paper standard.
3. Dataset: synchronized sensor traces, scale-validated sweat-loss references, randomization log, verified drink outcomes. De-identified and published if ethics permits. No such dataset exists publicly; the data-sharing clause must be in the week-1 ethics submission, as retrofitting consent is impossible.
4. Four-layer stack with ablation variants selectable by configuration flag.
5. Minimal dashboard.
6. One paper.

9. Reduced-Scope Fallback

If ethics or recruitment fails:

- Keep: grey-box deficit tracker, per-user calibration, full RQ1/RQ2 ablation, self-collected on 3–4 team members across 15–20 calibration sessions.
- Drop: MRT, receptivity evaluation, RQ3–RQ5.

- Result: a sensor-characterization paper — what does Rs. 2,600 of physiological sensing add to validated sweat-rate equations? — with real ground truth. Weaker than the full version, but strictly stronger than a system demo, because the label is measured rather than assumed.

If only one element of this plan can be built, build the bottle load cell. It converts every downstream claim from self-reported to measured.

10. Limitations to State in the Paper

1. Body-mass change as a sweat-loss reference carries known error from respiratory and metabolic mass loss and incomplete drying [6][7].
2. Wrist PPG during activity is heavily artifact-corrupted; the usable data fraction will be reported as a result, not hidden.
3. $n = 10-12$ over 14 days supports within-subject proximal effect estimation but not between-subject generalization.
4. An athletic cohort limits transfer to sedentary, elderly, or occupational populations.
5. Non-bottle drinking is captured only by manual log; bypass rate is a reported limitation.
6. The system estimates fluid balance, not hydration status, and makes no diagnostic claim.

11. Literature-Search Caveat

The prior-art assessment in Section 2 was assembled from web-accessible sources. **IEEE Xplore conference proceedings are underindexed by general web search, and that is where ESP32-based hydration prototypes cluster.** Before submission, run equivalent queries on IEEE Xplore, Scopus, and Google Scholar for: hydration JITAI and adaptive hydration intervention; heat-strain prompting and occupational heat wearables; micro-randomized trial combined with hydration (none located — confirm); and ESP32 hydration monitoring (expect many low-rigour conference papers, which still matter for related work). The judgement that G3–G5 are open should be verified against the embedded-systems literature before being written into an abstract.

12. Open Items Before External Circulation

1. Reference [12] requires bibliographic verification — journal and volume are unconfirmed. Do not cite until the record is pulled.
2. Full metadata for [2], [3], [4] and [17] should be checked against the publisher record; DOIs and PMIDs are confirmed but author lists were assembled from indexed summaries.
3. The constrained-bandit formulation in C2 is currently asserted rather than stated. Before submission, write the constrained optimization explicitly — objective, constraint, and the constraint variable's dynamics — or describe it plainly as a policy with a safety override.
4. If the JITAI-methods framing is adopted, RQ5 should move to the head of Section 5 and receive its own subsection in the analysis plan with the divergence metric formally defined.
5. Confirm access to 10–12 athletes for two weeks, and confirm the institutional ethics timeline for body-mass measurement and continuous physiological logging, before ordering parts.

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